



New 'category-defying' PC?

Corsair recently teased a fully assembled PC they call "The One". Pricing and specs are yet to be revealed. <http://digit.in/CorsairOne>

Fire TV boss joins Apple

Apple hires Timothy Twerdahl, the director of Amazon's Fire TV division, as a new vice president for their TV division. <http://digit.in/AppleTVboss>

actions would be delayed or denied in an event when the central server suffers downtime or gets compromised. This is how the peer-to-peer network operates ensuring that the database is never lost. Let's move on to talk about how the database is maintained.

In its initial state, generally called the genesis state, the entire network is notified and nodes are initialised. Now, transactions are broadcasted to the network so that the database can be modified with the required entries. These new entries aren't directly added but held up in a temporary state or pool consisting of one or more transactions. A finite collection of these pending transactions are called blocks. The transactions inside a block together generate a hash key or a digital signature unique to the particular block and

previous state. Because of the nature of the network, everyone will be able to notice the mismatch in the hash values and flag it off as fraudulent. As more transactions are broadcasted, more blocks will be created to form a chain, hence known as a blockchain.

If an attacker with a malicious intent wanted to modify any of the older blocks high up in the chain, it will be a foolish effort. Why? Well, this is another amazing security feature of a blockchain. You already know that modifying the transactions in a block to change the hash signature. You also know that the blocks are linked to the previous block using the hash signatures. If someone modified a block in between, the hash value of that block will change and break its link to the previous block. This will render all the

of blockchain is that this technology can be used to record or store any kind of information on a public audit or ledger system. Now, that you have a rough idea about its mechanism and why is there a need of blockchain, we will explore how this technology is going to enable sectors in the near future apart from what it was originally intended for – cryptocurrency.

Banking

Another sector where the use of blockchain is widely being promoted and funded is banking. Banks already have massive databases. Once they move to blockchain, banks will be able to process payments faster and more accurately. According to a report by Accenture, investment banks will be able to cut \$8 - \$12 billion a year by 2025 by introducing blockchain technology in their infrastructure. ICICI Bank was the first bank in the country to execute a transaction using blockchain last year. They exchanged and authenticated remittance messages and original international trade documents on a custom-made blockchain.

Recently, the State Bank of India launched a blockchain finance consortium called Bankchain. They've partnered with several technology firms and local commercial banks with technical assistance provided by IBM and Microsoft.

Music distribution

The challenges faced by the music industry is the inefficiency tracking music ownership and payments worldwide. Due to the lacking process of ownership and rights, it becomes difficult to manage the distribution of payments to everyone concerned with the project – a process taking years at times. By creating a new blockchain codec, the content will be added along with the information to the blockchain ready to be broadcasted.

Every time a new song is created by artists or record labels, it will be encoded with a blockchain codec and pushed onto the global database. From then on, the digital service providers



The whole network plays a part in authenticating blockchain transactions

set of transactions. With the Genesis state already existing, the new block should obviously denote the genesis state to be its previous state. The hash key exactly does the same by creating a link to the previous state. This linking is essential to the security of the transactions in the first block. The integrity of the first block is also maintained since if any changes were to be made to the transactions inside the block, the corresponding hash will change. This would result in a mismatch of the hash with the hash previously added in the

blocks following the newly modified block to become invalid. The nodes already store the database across the network and when the new block is broadcasted, a mismatch will be detected by the nodes. This is practically how a blockchain works.

Why do we need blockchain?

The simple answer is security. A distributed system is difficult to penetrate. For an intrusion attempt to be successful it'll have to be launched on all nodes simultaneously. Another beauty